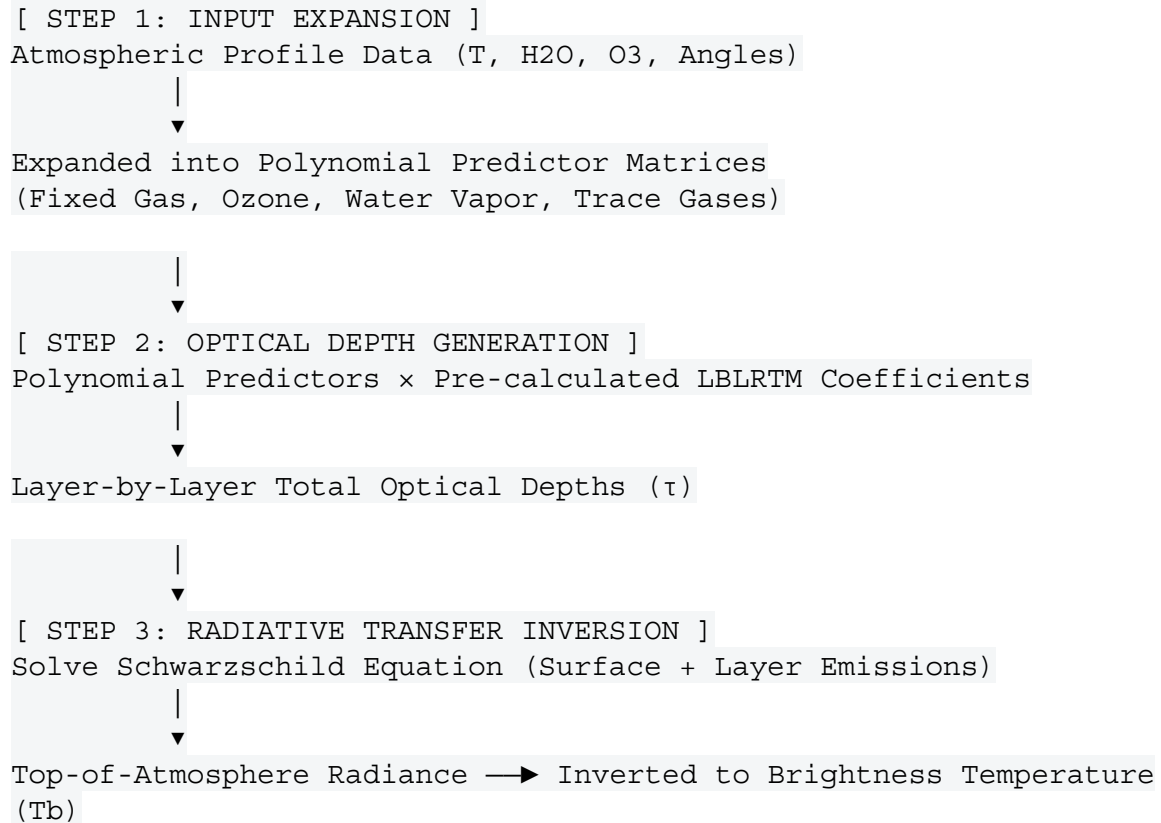


Yes, exactly! You have perfectly summarized the core engine of SARTA.

To map it out step-by-step, the data flows like a conveyor belt through three major stages:



The Three-Step Breakdown

1. **The Predictor Inputs:** SARTA takes standard physical inputs (a 100-layer profile of Temperature, H_2O , and O_3 from a weather model or satellite guess) and blows them up into those 10-to-15 term polynomial strings. This linearizes the complex, non-linear quantum physics of molecular absorption.
2. **Building the Optical Depths:** It takes those expanded predictor strings and multiplies them by its pre-calculated instrument coefficients. This instantly gives you the exact layer-by-layer optical thickness (τ) for every single satellite channel.
3. **The Radiative Transfer:** Armed with those optical depths, it runs the quick vertical integration (solving the Schwarzschild equation, adding surface emissions and reflections), and pops out the final Top-of-Atmosphere Radiance, which it converts directly into Brightness Temperature (T_b). [1]

By organizing the math this way, SARTA skips the millions of slow line-by-line integration loops that a model like LBLRTM has to do, reducing the calculation time down to mere milliseconds

per profile.

If you want to dig deeper into any specific part, I can show you how this forward model is flipped into an **inverse code (retrieval algorithm)** to calculate actual weather profiles, or we can look closer at the **Jacobian weight matrices** it outputs. Which path would you like to explore?

[1] <https://journals.ametsoc.org>